When Are BundleFuse's Scores Sufficient?

A theory of what to retain, what can be proved, and where to stop

Standalone analysis 7 November 2026

The question. BundleFuse can construct several scores from historical bundles and item attributes. Why retain these scores, and why not add another one? A satisfactory answer must explain the estimation problem, identify what an extra score could contribute, and specify a defensible stopping condition. Good benchmark results alone do not determine a necessary number of scores.

The conclusion. BundleFuse is a complete estimator whose final rule has an exact optimization interpretation. Its organizing idea is to start from relationships shared across items and permit departures supported by a candidate's history. The present score list is not universally sufficient: an explicit four-item construction defeats it. A rigorous, achievable target is a compact estimator that is within a stated ranking tolerance of a declared family of alternatives under a resource budget.

This document derives the relevant statements and distinguishes their status: an identity of the implemented estimator, a theorem under stated assumptions, a verified counterexample, or a proposed statistical certificate. The foundational sufficiency and shrinkage arguments are established mathematics; their application here does not establish that these principles are new.

1 The prediction problem and the source of the difficulty

Let $\mathcal{I}$ be a catalogue of n items. A query supplies a nonempty observed set $S \subset \mathcal{I}$. The eligible candidates are $\mathcal{I}_S = \mathcal{I} \setminus S$. We must rank them to recover a nonempty hidden set $G \subseteq \mathcal{I}_S$. The training collection contains m bundles, represented by the binary matrix

$$B \in \{0,1\}^{m \times n}, \qquad B_{bi} = 1 \iff \text{training bundle } b \text{ contains item } i. \tag{1}$$

Define the item frequency $f_i = \sum_b B_{bi}$ and the bundle size $d_b = \sum_i B_{bi}$. These definitions concern binary, unit-weight bundles, as in the fashion analysis below. Other data profiles may use weighted inputs.

Two shoes can share a category and have similar descriptions. Such shared attributes support a starting prediction for both. If only one repeatedly appears with the particular trousers and shirt in S, recorded item identities can distinguish them. For a newly introduced shoe, however, no historical match could mean either an unsuitable combination or simply an unobserved combination. The zero count alone cannot decide which.

This motivates one estimation problem: *use relationships that can be shared across items, then allow a candidate's specific observations to adjust the prediction.* A candidate with little recorded history should have a restricted ability to override that starting point. Candidate frequency is an available measure of exposure; it is not a calibrated measure of correctness, and frequent appearance does not imply frequent appearance with this query.

The different score constructions are possible ways of estimating these relationships. Their names do not imply independent information, and a count of score columns is not a count of independent observations.

2 One estimator: a shared prediction and a controlled departure

For a fixed candidate c and query S, let Y_c denote the selected full score and P_c the selected restricted fallback. Let $s > 0$ be the validation-frozen fallback scale, and define $F_c = sP_c$. The selected fallback strength is $\kappa \geq 0$. For $\kappa > 0$, BundleFuse computes

$$\boxed{\theta_c = F_c + \rho_c(Y_c - F_c), \qquad \rho_c = \frac{f_c}{f_c + \kappa}.} \tag{2}$$

All these scores depend on S; the query argument is suppressed for readability. The difference $Y_c - F_c$ is the proposed departure from the fallback. The coefficient ρ_c determines how much survives. For $f_c = 0$, the result is F_c. Holding Y_c and F_c fixed, the result approaches Y_c as f_c increases.

The fallback's information must be described accurately. Category relations pool different item IDs carrying the same label. Description scores compare attributes. The bridge also uses historical neighbors, and the fallback can include popularity. Thus F_c is neither an independent estimate nor an exclusively descriptive one. Since s need not equal one, the departure $Y_c - F_c$ need not contain only historical terms.

2.1 An exact optimization interpretation

Proposition 1 (The implemented interpolation)**.** *For fixed Y_c, F_c, f_c and $\kappa > 0$, equation (2) is the unique minimizer over real t of*

$$J_c(t) = \frac{f_c}{2}(t - Y_c)^2 + \frac{\kappa}{2}(t - F_c)^2. \tag{3}$$

Proof. The derivative is $J_c'(t) = (f_c + \kappa)t - f_c Y_c - \kappa F_c$. Its zero gives equation (2). The second derivative is $f_c + \kappa > 0$, so the minimizer is unique. □

The two terms charge for departing from the full score and from the fallback, respectively. The first charge grows with historical exposure. This interpretation needs no probability model for the scores. It does not derive the raw score menu or establish that its weights are optimal.

When $\kappa = 0$, the implementation returns Y_c. If also $f_c = 0$, the objective in (3) is constant and does not uniquely specify that answer. The explicit convention is therefore part of the estimator.

There is also an exact stability property. If Y_c changes by e_c while F_c stays fixed, θ_c changes by $\rho_c e_c$. For candidates c and d, the change in their score difference is at most

$$\rho_c|e_c| + \rho_d|e_d|. \tag{4}$$

This limits sensitivity to changes confined to the full score, especially at low frequency. It does not guarantee better rankings: a suppressed departure could have been correct. If a new source changes the fallback too, that change must also be included.

2.2 What the dense model actually selects

Let $r_h(c \mid S)$ be a raw score with source index h. In the examined full-ranking implementation it is standardized over all n catalogue items before observed items are masked:

$$\mu_h(S) = \frac{1}{n}\sum_i r_h(i \mid S), \qquad \sigma_h(S)^2 = \frac{1}{n-1}\sum_i [r_h(i \mid S) - \mu_h(S)]^2,$$
$$z_h(c \mid S) = \frac{r_h(c \mid S) - \mu_h(S)}{\sigma_h(S) + 10^{-9}}. \tag{5}$$

The candidate comparisons use different functions of the same available data. Standardization balances their numerical scales; it does not establish independence or calibrate them as probabilities.

For the frozen POG_DENSE research recipe, define $Q_c = P_c$. The selected construction is

$$\begin{aligned} Q_c &= 0.5z_{\text{cat}}(c) + 0.25z_{\text{bridge}}(c) + 0.25z_{\text{pop}}(c), \\ Y_c &= Q_c + 0.5z_{\text{simp}}(c) + 0.25z_{\text{csvd}}(c), \\ F_c &= sQ_c, \qquad s \simeq 1.46055159, \qquad \kappa = 0.5, \\ \theta_c &= [s + (1-s)\rho_c]Q_c + \rho_c[0.5z_{\text{simp}}(c) + 0.25z_{\text{csvd}}(c)]. \end{aligned} \tag{6}$$

Selected score	Relationship being measured
cat	Training relations between item categories, with a penalty for repeating categories that rarely repeat within a bundle.
bridge	Similarity between category-centered descriptors after incorporating descriptions of historical neighbors.
pop	The candidate's log training frequency, $\log(1 + f_c)$.
simp	Candidate support from individual query items and from query-item pairs appearing together with it.
csvd	Similarity between compressed profiles of item co-occurrence associations.

Appendix A defines these five raw scores mathematically. The dense recipe does not select standalone cooc, ppmi, soft-group, propagated-score or Rao columns. PPMI and propagation still appear inside selected representations; their absence as separate columns does not remove those internal operations. The frozen POG recipe uses four columns; the Spotify recipe uses seven weighted columns plus its explicit Rao term. None of these counts has been proved necessary or minimal.

3 What information is sufficient for a ranking?

A sufficiency claim must identify the decision and the population. Let X contain all permitted prediction-time information: S, the training data, item attributes, candidate eligibility and any allowed user context. The hidden set G is random under the specified population of queries. Let $K \le |\mathcal{I}_S|$ be the cutoff and let $\pi(j)$ be the candidate at position j.

For Recall@K, define the candidate contribution V_c and position weight ω_j by

$$V_c = \frac{\mathbf{1}\{c \in G\}}{|G|}, \qquad \omega_j = 1 \quad (1 \le j \le K). \tag{7}$$

For binary-relevance NDCG@K, instead define

$$\omega_j = \frac{1}{\log_2(j+1)}, \qquad D_K(G) = \sum_{j=1}^{\min(K,|G|)} \omega_j, \qquad V_c = \frac{\mathbf{1}\{c \in G\}}{D_K(G)}. \tag{8}$$

Here $D_K(G)$ is the largest attainable discounted gain for that hidden set. In either case, realized utility equals $\sum_{j=1}^{K} \omega_j V_{\pi(j)}$. Define the expected contribution

$$\eta_c(X) = \mathbb{E}[V_c \mid X]. \tag{9}$$

The conditional expected utility of a ranking is therefore

$$U(\pi \mid X) = \sum_{j=1}^{K} \omega_j \eta_{\pi(j)}(X). \tag{10}$$

Sorting η maximizes this expression. To see this, place candidates c and d at positions $j < k$, where $\omega_j \geq \omega_k$. If $\eta_c < \eta_d$, exchanging them improves utility by $(\omega_j - \omega_k)(\eta_d - \eta_c) \geq 0$. Treat positions below K as having weight zero. Repeated exchanges produce a sorted ranking. For Recall only the selected set matters; for NDCG its order matters too.

Thus a representation need only preserve an optimal ordering, rather than the entire distribution of G. When $|G|$ varies, plain membership probabilities need not have the same ordering as these expected normalized contributions. The NDCG distinction is established in the consistency analysis of Ravikumar, Tewari and Yang [1].

BundleFuse's numerical scores are not calibrated estimates of η. Consequently, errors between its raw numerical θ and η cannot be interpreted as calibrated prediction errors without further justification. Only their rankings are directly compared in the existing evaluation.

4 The value of one more score

Let Z retain the full vectors of selected candidate scores, frequencies and eligibility information. Let R be a proposed additional score vector. These are functions of allowed inputs, not hidden labels. For the contribution vector V defined for the chosen metric, write

$$\begin{aligned} m_0(Z) &= \mathbb{E}[V \mid Z], & m_1(Z,R) &= \mathbb{E}[V \mid Z,R], \\ \Delta(R \mid Z) &= \mathbb{E}\big[\, \|m_1 - m_0\|_2^2 \,\big]. \end{aligned} \tag{11}$$

The norm sums over eligible candidates. Expectations refer to the declared query population. Δ measures how much the best prediction of the relevant items changes after observing R. It is not a correlation between raw scores and not the standalone performance of R.

Proposition 2 (Exact decrease in optimal prediction error)**.**

$$\mathbb{E}\big[\, \|V - m_0\|_2^2 \,\big] - \mathbb{E}\big[\, \|V - m_1\|_2^2 \,\big] = \Delta(R \mid Z). \tag{12}$$

Proof. Expand $V - m_0 = (V - m_1) + (m_1 - m_0)$. Conditional on (Z, R), $V - m_1$ has mean zero, while $m_1 - m_0$ is fixed. The expected cross term is therefore zero. Expanding the squared norm gives the identity. □

If $\Delta = 0$, revealing R cannot improve optimal expected ranking utility for this metric: the two conditional mean vectors are equal almost surely. A positive Δ need not improve utility, because changed values can retain the same top-K ordering. This result uses the classical projection property of conditional expectation [2]. Its relation to statistical sufficiency is that relevant prediction information is preserved; full conditional independence is a stronger condition [3].

Proposition 3 (Bound on additional ranking utility)**.** *Let $\boldsymbol{\omega} = (\omega_1, \ldots, \omega_K)$. The improvement in optimal expected utility from revealing R is at most*

$$\min\left\{1,\ 2\,\|\boldsymbol{\omega}\|_2 \sqrt{\Delta(R \mid Z)}\right\}. \tag{13}$$

Proof. Represent a ranking by a vector a assigning weight ω_j to its jth candidate and zero below K. Every ranking vector has norm $\|\boldsymbol{\omega}\|_2$. Let a_0 maximize $a^\top m_0$ and a_1 maximize $a^\top m_1$. Then

$$\begin{aligned} (a_1 - a_0)^\top m_1 &= (a_1 - a_0)^\top m_0 + (a_1 - a_0)^\top (m_1 - m_0) \\ &\leq 2\,\|\boldsymbol{\omega}\|_2\, \|m_1 - m_0\|_2 \,. \end{aligned} \tag{14}$$

The first term is nonpositive by optimality of a_0; the second is bounded by the Cauchy–Schwarz and triangle inequalities. Take expectations and use $\mathbb{E}[\|W\|_2] \leq \sqrt{\mathbb{E}[\|W\|_2^2]}$. Both evaluation metrics lie in $[0, 1]$, which supplies the cap of one. □

For a specified class of admissible extensions, a population-level stopping condition is therefore

$$\sup_R \Delta(R \mid Z) \leq \frac{\varepsilon^2}{4 \|\boldsymbol{\omega}\|_2^2}. \tag{15}$$

Under this condition, no admissible extension improves the best Z-based ranking by more than ε. The class and population must be specified; the inequality is not known to hold for the present BundleFuse scores. It cannot simply be assumed as a proof of their sufficiency.

4.1 Existing information can also be used better

The current combiner might not extract all the information in Z. Let $h(Z)$ be any numerical prediction on the same scale as V, and define $E_0 = \mathbb{E}[\|h - m_0\|_2^2]$. A second orthogonality identity gives

$$\mathbb{E}[\|h - m_1\|_2^2] = E_0 + \Delta(R \mid Z). \tag{16}$$

Indeed, $\mathbb{E}[m_1 - m_0 \mid Z] = 0$, making its expected inner product with $h - m_0$ zero. The same ranking argument bounds the gain over the ranking of h by $2 \|\boldsymbol{\omega}\|_2 \sqrt{E_0 + \Delta}$, capped at one. A calibrated h with BundleFuse's ranking has not been established, so this is a conditional interpretation, not a measured certificate for θ.

A new model can thus help by extracting existing information better, even when it adds no information. PPMI and SVD being derived from the same matrix does not make their score columns interchangeable for a restricted weighted combiner.

These results also apply when R jointly contains several new scores. Testing additions individually cannot rule out joint benefit. Two independent fair bits each reveal nothing about their exclusive-or, while together they determine it. Likewise, a nonsignificant addition does not prove $\Delta = 0$; the experiment may simply be too imprecise.

5 Why the present score menu is not universally complete

An explicit construction shows what the scores can miss. Create four item groups A, B, C, D, each containing q values indexed $0, \ldots, q-1$. For a fixed shift $h \in \{0, \ldots, q-1\}$, include every training bundle

$$\{A_a, B_b, C_c, D_{(a+b+c+h) \bmod q}\}, \qquad a, b, c = 0, \ldots, q-1. \tag{17}$$

There are q^3 bundles, each of size four. Use the group as the category label and give all items within each group the same descriptor.

Fixing r values from distinct groups, for $r = 1, 2, 3$, leaves exactly q^{3-r} possible bundles, independently of h. A subset containing two different values from one group occurs zero times. Hence complete single-item, pair and triple counts are identical for all shifts. All bundle size discounts are identical as well.

For $S = \{A_0, B_0, C_0\}$, however, the recorded completion is D_h. The four-item count identifies it immediately. With $q = 2$, the two possible shifts give the particularly small example

Observed triple	Completion when $h = 0$	Completion when $h = 1$
$\{A_0, B_0, C_0\}$	D_0	D_1

The failure of the examined fashion scores is specific:

- Direct pair, co-occurrence and PPMI scores use identical frequency and pair statistics. PPMI-SVD receives the same matrix, with identical association-profile rows for all D candidates.
- The incidence correction uses candidate–seed–seed triple counts, which are also identical. Its derivation appears in Appendix A.1.
- Propagation uses the same pair operator. Group-identical descriptors cannot distinguish D candidates; their category-centered residuals are zero.
- Category, soft-group and popularity scores cannot distinguish these candidates either. Rao uses the same candidate–seed affinities and the same relations among the query items.

Consequently, standardization and frequency interpolation receive equal inputs for every D value and produce equal scores in ideal arithmetic. An ID-based tie-break does not reconstruct the four-item relation. With $q = 21$, more D candidates are tied than there are top-20 positions. Varying h makes any one of them the correct answer while preserving the score inputs, so no fixed top-20 rule from these scores succeeds for every corpus. The ordering limitation for NDCG already appears with two values.

The construction was evaluated with the production fashion score operators and the frozen dense fusion for $q = 2$ and $q = 21$. It verified equal counts through order three and score equality to floating-point tolerance. The largest score differences between the two checked shifts were 2.39×10^{-7} and 1.20×10^{-7}, respectively. These are synthetic mathematical checks, not benchmark performance results.

The conclusion is limited but decisive: a theorem that this menu captures every possible completion relationship is false. It does not follow that such four-item relationships are frequent or useful in the intended data. That becomes an empirical question with a precise missing relationship to test, rather than an invitation to add arbitrary scores indefinitely.

6 A stopping rule with a finite statistical guarantee

Condition (15) is conceptually useful but generally difficult to establish. A more practical standard compares a finite, predeclared family of fitted systems under the ranking metric itself.

Let $\mathcal{H}$ contain M systems whose fitted parameters and procedures are frozen before a fresh audit sample is examined. Include the compact candidate, relevant source removals, close replacements and justified additions. Let $C(h)$ be a defined cost, such as inference time under a fixed measurement protocol. Memory and fitting cost can be separate hard constraints. Let $\varepsilon \geq 0$ be the acceptable absolute utility loss.

Suppose the audit contains N independent units from the target population. Each system's utility on each unit lies in $[0, 1]$. A unit can be a complete original bundle with a predefined average across its query masks. Repeated masks of the same bundle must not be counted as independent observations. User or temporal dependence can require a different unit or probability bound. Let $U(h)$ be population utility and $\widehat{U}(h)$ its audit mean. Condition on the independently fitted systems when applying the following argument.

For a failure probability $\delta \in (0, 1)$, define

$$b = \sqrt{\frac{\log(2M/\delta)}{2N}}. \tag{18}$$

Hoeffding's inequality [4] and a union bound imply

$$\Pr\left\{\text{for every } h \in \mathcal{H},\ |\widehat{U}(h) - U(h)| \leq b\right\} \geq 1 - \delta. \tag{19}$$

Proposition 4 (A compact-system guarantee)**.** *Choose a least-cost member* h_{small} *satisfying*

$$\widehat{U}(h_{\text{small}}) \geq \max_{h \in \mathcal{H}} \widehat{U}(h) - \varepsilon. \tag{20}$$

With probability at least $1 - \delta$*,*

$$U(h_{\text{small}}) \geq \max_{h \in \mathcal{H}} U(h) - \varepsilon - 2b. \tag{21}$$

Proof. On the simultaneous event, the selected system's population utility is at least its audit utility minus b. Its audit utility is at least the best audit utility minus ε. The best audit utility is at least the population-best member's utility minus b. Combine the inequalities. □

Simultaneity permits selecting among these frozen systems after seeing the audit means. It does not permit fitting new systems on the audit labels and pretending the original guarantee still applies. A particular baseline is certified within tolerance ε of the family if

$$\max_{h \in \mathcal{H}} [\widehat{U}(h) - \widehat{U}(\text{baseline})] + 2b \leq \varepsilon. \tag{22}$$

The elementary bound is often loose. With $M = 100$, $N = 4{,}000$ and $\delta = 0.05$, $2b \simeq 0.0644$ absolute utility. That exceeds several observed differences in this project. A narrow certificate would need a justified tighter paired bound or more independent data. A nonsignificant comparison is not a substitute for such a bound.

This defines adequacy relative to a population, metric, tolerance, resource constraint and comparison family. It does not cover an algorithm invented after the audit. It does provide a reason to stop: retain a compact system once justified alternatives cannot offer enough gain to warrant their cost.

7 What can be claimed about BundleFuse now?

The completed experiments support particular choices rather than an indispensable universal menu. The controlled incidence bank improves dense POG NDCG over its pairwise counterpart. On POG, history and shared relations work together and the fallback helps. On Spotify, a different historical overlap estimator improved the controlled result. Clothing and sparse Spotify discarded the explicit historical column in the follow-up. These findings favor a compact, dataset-selected implementation.

They do not establish the certificate in Section 6. The studies are exploratory supplied-split comparisons, conditional on earlier source inventories, and their test outcomes have informed the research. They did not search every subset or audit all relevant joint additions on fresh data.

A further caution concerns the normal observation model derived in Appendix B. It provides a real sufficient-aggregate theorem under its assumptions, but those assumptions are not established for the current standardized scores. The saved full recipes fit the mean squared $(Y - F)$ against inverse frequency and obtain these raw slopes:

Recipe	Raw inverse-frequency slope	Selected κ
POG	−0.273238	0.5
POG_DENSE	−2.528107	0.5
Spotify	−5.335897	5.0

These estimates do not substantiate a positive inverse-frequency noise relationship. They are not themselves formal rejection tests. The procedure truncates inadmissible variance estimates and considers validation-selected alternatives. Thus equation (3) is an unconditional account of the current interpolation; a calibrated posterior is not established.

7.1 A publication statement with a definite object of study

> BundleFuse estimates how an item completes an observed set from recorded item combinations and relationships shared across items. Its historical coefficients are determined by bundle counts. Its final rule limits departures from a shared prediction according to candidate history. The scorer requires no gradient-trained scoring network; its scalar choices are selected on validation data.

The exact historical estimator, the chosen sharing operations, their computation, and their controlled empirical effects are the objects to defend. The supplied item features can have upstream training costs; SVD, clustering and scalar selection are also fitted operations. "No gradient-trained scoring network" is more precise than "nothing is trained."

The reported comparisons with published CLHE/RaMen values support empirical effectiveness under the project's documented comparison scope. They do not prove score sufficiency, identify the cause of the advantage, or establish priority for the full construction. RaMen's task is bundle construction [9]. Related bundle recommendation already uses frequency-dependent mixtures, for example CoHeat [8], although its task and learned representations differ. A mathematical property and a claim of originality require separate evidence.

The response to "why not add another model?" should be precise:

> BundleFuse specifies a complete estimator with a limited fitting and computational budget. Retained components must justify their incremental ranking value. An extension is relevant when it addresses a missing relationship or improves the accuracy–cost tradeoff. The contribution does not depend on claiming that no future extension can ever help.

The present analysis establishes exact behavior and representational limits. It supplies a route to a compact-system guarantee. It does not certify that the current number of scores is minimal, or turn established sufficiency theorems into new algorithmic contributions.

A Definitions of the five dense score columns

This appendix makes the dense example self-contained. It defines the mathematical operations used in equation (6). Apart from the standardization stabilizer in (5), tiny numerical stabilizers in logs and row norms are omitted for readability. A zero vector is mapped to zero by row normalization. No proof here depends on those floating-point conventions.

A.1 Individual-seed and joint-seed historical support

Define the seed weight and bundle weight

$$a_i = (f_i + 5)^{-1/2}, \qquad v_b = (d_b + 1)^{-1/2}. \tag{23}$$

For a query, let $o_b(S) = \sum_{s \in S} B_{bs} a_s$ be its weighted overlap with training bundle b, and $k_b(S) = \sum_{s \in S} B_{bs}$ its number of overlapping items. The direct and additional scores are

$$\begin{aligned} p_c(S) &= \sum_b B_{bc} v_b o_b(S), \\ \delta_c(S) &= \beta \sum_b B_{bc} v_b o_b(S)[k_b(S) - 1]_+, \qquad \beta = 1.5, \\ g_c(S) &= p_c(S) + \delta_c(S), \qquad r_{\text{simp}}(c \mid S) = g_c(S), \end{aligned} \tag{24}$$

where $[x]_+ = \max(x, 0)$. Thus g_c and δ_c are different: the former includes the direct score.

Define the weighted triple count for distinct items

$$T_{cst} = \sum_b v_b B_{bc} B_{bs} B_{bt}. \tag{25}$$

Then

$$\delta_c(S) = \beta \sum_{\{s,t\} \subseteq S} (a_s + a_t) T_{cst}. \tag{26}$$

To verify this, a bundle with k query items contributes each seed weight once for each of the other $k - 1$ query items. Grouping the ordered contributions into unordered pairs gives $a_s + a_t$. For zero overlap, $o_b = 0$, so the same expansion remains valid.

The raw g_c is therefore quadratic in the binary query indicators, with coefficients tied to observed counts rather than fitted separately. It does not span every quadratic function. Nor does the whole final pipeline remain quadratic: query standardization and, in other selected configurations, the Rao denominator introduce other nonlinear dependence.

A.2 Category relations and repeated-category penalty

Let $\ell_i \in \{1, \ldots, L\}$ be the supplied category label of item i. The membership matrix $A \in \{0, 1\}^{n \times L}$ has $A_{i\ell} = \mathbf{1}\{\ell_i = \ell\}$. Define

$$H = BA, \qquad h_\ell = \sum_b H_{b\ell}, \qquad H_+ = \sum_\ell h_\ell, \qquad C^{\text{cat}} = \text{off}(H^\top H). \tag{27}$$

$H_{b\ell}$ counts items of category ℓ in bundle b; off sets the diagonal of a square matrix to zero. For a positive off-diagonal count, define

$$P^{\text{cat}}_{\ell r} = \max\left(0, \log \frac{C^{\text{cat}}_{\ell r} H_+}{h_\ell h_r}\right), \tag{28}$$

and set it to zero when the count is zero or $\ell = r$. Let

$$\eta_\ell = \frac{\sum_b \mathbf{1}\{H_{b\ell} \geq 2\} + 0.5}{\sum_b \mathbf{1}\{H_{b\ell} \geq 1\} + 1}. \tag{29}$$

This is a smoothed empirical repetition rate, not a supplied semantic rule. The fashion category score is

$$r_{\text{cat}}(c \mid S) = \sum_{s \in S} \left[P^{\text{cat}}_{\ell_s \ell_c} - \frac{1}{2} \mathbf{1}\{\ell_s = \ell_c\} \mathbf{1}\{\eta_{\ell_s} < 0.15\} (1 - \eta_{\ell_s}) \right]. \tag{30}$$

It pools category-pair observations and penalizes repetition of categories that rarely repeat in training. All candidates in one category receive the same value for a fixed query.

A.3 Descriptions combined with historical-neighbor descriptions

For each available description or image feature source u, let $x_i^{(u)}$ be its fixed, normalized input vector. These are supplied representations; this scorer does not train their upstream encoders. For each category ℓ, define its arithmetic mean

$$\bar{x}_\ell^{(u)} = \frac{1}{|\{i : \ell_i = \ell\}|} \sum_{i:\ell_i=\ell} x_i^{(u)}. \tag{31}$$

Subtract this mean from each item's vector and normalize the result to unit length when nonzero. Concatenate these residual vectors across available sources and normalize again, producing row vectors e_i. Let E be the matrix with rows $e_i^\top$.

Define $D = \mathrm{diag}(a_i)$ and the symmetric historical propagation matrix

$$L_B = \mathrm{off}\left[DB^\top \mathrm{diag}\left((d_b+1)^{-1}\right)BD\right]. \tag{32}$$

This is an adjacency operator, not a Laplacian or a row-stochastic transition matrix. Let rownorm normalize each nonzero row to unit Euclidean length. The bridge representation and score are

$$W = \mathrm{rownorm}\left(E + \tfrac{1}{2}\,\mathrm{rownorm}(L_B E)\right), \qquad r_{\mathrm{bridge}}(c \mid S) = w_c^\top \frac{1}{|S|}\sum_{s\in S} w_s, \tag{33}$$

where $w_i^\top$ is row i of W. It compares an item's own residual description together with an aggregate of its neighbors' residual descriptions. This explains why the bridge includes historical information.

A.4 Co-occurrence profiles compressed by SVD

Define the item-pair counts and their endpoint totals

$$C = \mathrm{off}(B^\top B), \qquad e_i^{\mathrm{cnt}} = \sum_j C_{ij}, \qquad C_+ = \sum_{ij} C_{ij}. \tag{34}$$

The endpoint totals differ from item frequency f_i: appearances in larger bundles contribute more partners. For $C_{ij} > 0$, define the positive pointwise mutual information matrix

$$P_{ij} = \max\left(0, \log \frac{C_{ij} C_+}{e_i^{\mathrm{cnt}} e_j^{\mathrm{cnt}}}\right), \tag{35}$$

and set $P_{ij} = 0$ otherwise. Let V_r contain retained right singular vectors in a rank-r SVD of P, and form the item representation $X_{\mathrm{svd}} = PV_r = U_r\Sigma_r$ for an exact decomposition. The implementation uses a truncated numerical approximation. Normalize the rows of X_{svd} to obtain u_i. Then

$$r_{\mathrm{csvd}}(c \mid S) = u_c^\top \frac{1}{|S|}\sum_{s\in S} u_s. \tag{36}$$

This compares compressed association profiles. It does not directly reconstruct the PPMI entry between each candidate and seed. Orthonormal singular-vector columns do not imply unit-length rows of $U_r\Sigma_r$, which is why row normalization is a separate operation. The representation principle has explicit precedents [6, 7].

A.5 Popularity and selection

The remaining raw score is $r_{\mathrm{pop}}(c \mid S) = \log(1+f_c)$, independent of the query before standardization. Apply (5) to each of the five raw columns, then use (6). The displayed constants describe the frozen recipe; these formulas do not prove that those constants or exactly these five columns are optimal.

B A sufficient aggregate under correlated normal measurements

This appendix gives a conditional theorem connecting several correlated measurements to a single aggregate and to equation (2). It is an explanatory model, not a distributional claim about the current raw scores or about binary hidden-item indicators.

For one candidate and query, let t be an unknown real compatibility quantity. Condition on the information used to construct a fixed fallback F and on frequency $f > 0$. Suppose a p-dimensional vector of bias-corrected measurements $\mathbf{u}$ obeys

$$\mathbf{u} = \boldsymbol{\ell}\, t + \boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \mid t, F, f \sim \mathcal{N}(0, \Sigma/f), \qquad t \mid F, f \sim \mathcal{N}(F, \tau^2). \tag{37}$$

The response vector $\boldsymbol{\ell}$, positive-definite covariance Σ, and prior variance $\tau^2 > 0$ are known within this conditional model. Error correlations are allowed. Define

$$I = \boldsymbol{\ell}^\top \Sigma^{-1} \boldsymbol{\ell} > 0, \qquad y = \frac{\boldsymbol{\ell}^\top \Sigma^{-1} \mathbf{u}}{I}, \qquad \kappa = \frac{1}{\tau^2 I}. \tag{38}$$

Proposition 5 (Conditional sufficient aggregate)**.** *For known $\boldsymbol{\ell}, \Sigma, f$, y is sufficient for t in this likelihood, and*

$$\mathbb{E}[t \mid \mathbf{u}, F, f] = F + \frac{f}{f + \kappa}(y - F). \tag{39}$$

Proof. The part of the negative log-likelihood depending on t is

$$\frac{f}{2}(\mathbf{u} - \boldsymbol{\ell} t)^\top \Sigma^{-1} (\mathbf{u} - \boldsymbol{\ell} t) = \frac{fI}{2}(t - y)^2 + \text{constant in } t. \tag{40}$$

The data enter through y. Adding the prior penalty $(t - F)^2/(2\tau^2)$ and completing the square gives the posterior mean. This is standard normal conjugacy [5]. □

The theorem permits summarizing a specified measurement vector. It does not say that a new measurement can never help. Suppose a new measurement has response coefficient ℓ_{new}, error variance v/f, and cross-covariance k/f with the old errors. Assume the expanded errors are jointly normal with a positive-definite covariance and remain independent of t conditional on F, f. Define

$$\ell_\perp = \ell_{\text{new}} - k^\top \Sigma^{-1} \boldsymbol{\ell}, \qquad v_\perp = v - k^\top \Sigma^{-1} k > 0. \tag{41}$$

Subtracting $k^\top \Sigma^{-1} \mathbf{u}$ from the new measurement removes the part predictable from old errors. The remainder has response $\ell_\perp$ to t and independent normal error variance $v_\perp/f$. Therefore the aggregate precision factor becomes

$$\boxed{I_{\text{new}} = I + \frac{\ell_\perp^2}{v_\perp}.} \tag{42}$$

The additional measurement contributes no information exactly when $\ell_\perp = 0$ under this model. A nearly duplicated measurement can still help if its remaining part measures t accurately. Exact duplicates give a singular expanded covariance and should be removed before using these inverse formulas.

For current BundleFuse, the required assumptions are unverified. Its grid weights are not these covariance-adjusted coefficients. Its full score and fallback share inputs. A historical sum is not automatically a sample mean whose error variance is proportional to $1/f$. The distributional parameters are not known. A single global κ would additionally require $\tau^2 I$ to be constant across candidate/query conditions. The negative diagnostic slopes in Section 7 reinforce the need to keep this conditional theorem separate from an unconditional claim about the implemented scores.

C Evidence and reproducibility

The mathematical checks are deliberately small and synthetic. They neither fit new benchmark models nor alter existing checkpoints. Run from the repository root:

```
PYTHONPATH=.  python scripts/check_score_sufficiency.py
```

The retained result is `results/algebraic_diagnostics/score_sufficiency.json`. It records source hashes and the following checks:

- The $q = 2$ construction: 8 items and 8 bundles per corpus.
- The $q = 21$ construction: 84 items and 9,261 bundles per corpus.
- Equal frequency, pair and triple counts; equal fashion score vectors and frozen dense fusion to floating-point tolerance; different four-item completion counts.
- The precision-increment identity in 100 positive-definite covariance cases, with maximum numerical discrepancy 2.78×10^{-15}.
- The expected normalized-relevance ranking rule and additional-utility bound in 100 Recall and 100 NDCG cases, enumerating all 24 rankings of four candidates. Maximum discrepancy from the optimal sorted utility is 2.23×10^{-16}.

The checks illustrate and audit the derivations; the proofs, rather than the finite numerical cases, establish the propositions.

The frozen recipes and variance diagnostics are stored in `results/publication_mechanisms/pog/full.json`, `results/publication_mechanisms/pog_dense/full.json`, and `results/publication_mechanisms/spotify/full.json`. The completed controlled studies are recorded under `results/novelty_attribution/` and `results/amplified_fallback/`. Those experiments support the empirical statements in Section 7; they are not the independent audit required by Section 6.

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